Interconnections

Signals first, fractions second

A feedback diagram is a set of simultaneous equations, not a recipe for multiplying transfer functions around a circle. First decide who owns each signal. Then ask whether the equations define a causal system. Only then ask whether its stored energy decays.

> The route: cut the controller out → assign ports → solve the instantaneous loop → eliminate signals → test hidden dynamics → ask whether stabilization is possible. The final three sections solve all three book exercises, including a decentralized design.

1 Cut the controller out and assign ownership

The open-loop interconnection is everything left after removing the controller, not necessarily a physical plant alone. It includes sensors, weights, summing junctions, and fixed feedback paths. "Open" means open at the chosen controller ports; fixed loops inside it must already be well posed. Chapter 3 of [1] organizes these potentially complicated equations around four signals:

w	External causes: references, disturbances, noise. Not selected by the controller.
u	Commands: the controller's outputs, treated as independent inputs while the loop is cut.
y	Available measurements: the controller's inputs, not every state or performance variable.
z	Performance variables: what the designer wants small; not automatically measured.

This ownership matters: feeding back z when only y is available changes the problem. Stacking several scalar signals makes a vector; scalar component blocks do not make the overall interconnection SISO.

The tracking loop as the running example [1, pp. 46–48]. A plant G is controlled by $u = Ky$ with $y = r - (Gu + d) - n$, and the tracking error is $e = (Gu + d) - r$: output minus reference, unlike chapter 2's measured error. Sort the signals by role: exogenous $w = (d, n, r)$, which the controller cannot influence; the controlled output $z = e$, which defines success; the control input u; the measurement y. With the controller removed, the loop is a plain linear map

$$\begin{pmatrix} e \\ y \end{pmatrix} = \begin{pmatrix} I & 0 & -I & G \\ -I & -I & I & -G \end{pmatrix} \begin{pmatrix} d \\ n \\ r \\ u \end{pmatrix}, \tag{1}$$

so $P_{11} = \begin{pmatrix} I & 0 & -I \end{pmatrix}$, $P_{12} = G$, $P_{21} = \begin{pmatrix} -I & -I & I \end{pmatrix}$, $P_{22} = -G$. Closing $u = Ky$ and simplifying yields the familiar $e = \begin{pmatrix} S & -T & -S \end{pmatrix} \begin{pmatrix} d \\ n \\ r \end{pmatrix}$ with $S = (I + GK)^{-1}$, $T = GK(I + GK)^{-1}$. Every other block diagram in robust control is processed the same way.

> The convention is **positive interconnection** $u = Ky$. Ordinary negative tracking feedback is already encoded in $y = r - Gu - d - n$, hence $P_{22} = -G$ and the familiar denominator is $I + GK$. Do not add another minus sign. Even here, checking only S is not an internal-stability test without noncancellation assumptions.

2 Can the instantaneous equations be solved?

The general framework [1, pp. 48–50]. Any interconnection, once the controller is cut out, reads

$$\begin{pmatrix} z \\ y \end{pmatrix} = P \begin{pmatrix} w \\ u \end{pmatrix} = \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix} \begin{pmatrix} w \\ u \end{pmatrix}, \quad u = Ky, \tag{2}$$

with a stabilizable and detectable realization

$$\dot{x} = Ax + B_1 w + B_2 u, \quad z = C_1 x + D_{11} w + D_{12} u, \quad y = C_2 x + D_{21} w + D_{22} u, \tag{3}$$

and a controller $\dot{x}_K = A_K x_K + B_K u_K, y_K = C_K x_K + D_K u_K$, also stabilizable and detectable. Even a loop of SISO blocks gives a MIMO P, because signals are stacked.

These assumptions concern **all** inputs (w, u) and outputs (z, y). They allow extra stable hidden states, but no unstable uncontrollable or unobservable states. Minimal (controllable and observable) realizations suffice and simplify computation. Never silently replace a physical realization by its minimal transfer realization when checking physical internal stability: cancellation could discard the very mode at issue.

At a fixed instant the states are already known. Write $u_0 = C_K x_K$ and $y_0 = C_2 x + D_{21} w$. The unknown instantaneous signals solve

$$u = u_0 + D_K y, \quad y = y_0 + D_{22} u, \tag{4}$$

$$\begin{pmatrix} I & -D_K \\ -D_{22} & I \end{pmatrix} \begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} u_0 \\ y_0 \end{pmatrix}. \tag{5}$$

Unique solutions for all states and inputs require

$$M := \begin{pmatrix} I & -D_K \\ -D_{22} & I \end{pmatrix} \text{ nonsingular.} \tag{6}$$

Equivalently $I - D_{22} D_K$ is nonsingular (Schur complement). This is **algebraic well-posedness**, not a statement about decay over time. For proper rational P, K it is equivalent to $I - P_{22} K$ having a proper rational inverse.

Two contrasting checks. If $y = w + 2u$ and $u = \frac{y}{2}$, then $y = w + y$: no solution for $w \neq 0$, infinitely many for $w = 0$. This is not merely an unstable system; it fails to define an ordinary causal system. If instead $\dot{x} = x + u, y = x, u = 0$, then $D_{22} = 0$ and the loop is perfectly well posed, yet $x(t) = \exp(t)x(0)$ grows. When $D_{22} = 0$, every proper controller is well posed; stability remains a separate task.

3 Eliminate signals, then name the lower LFT

Start with the signal equations, at zero initial state:

$$z = P_{11} w + P_{12} u, \quad y = P_{21} w + P_{22} u, \quad u = Ky. \tag{7}$$

Substitute the third into the second, solve, and substitute into the first:

$$(I - P_{22} K) y = P_{21} w, \quad y = (I - P_{22} K)^{-1} P_{21} w, \tag{8}$$

$$u = K(I - P_{22} K)^{-1} P_{21} w, \tag{9}$$

$$z = \left(P_{11} + P_{12} K (I - P_{22} K)^{-1} P_{21} \right) w. \tag{10}$$

Only now introduce a name for this result:

$$P \star K := P_{11} + P_{12} K (I - P_{22} K)^{-1} P_{21}. \tag{11}$$

This is the **lower linear fractional transformation**, also written $F_{l(P,K)}$. It is a compact name for elimination, not a new physical operation. For the tracking equations it gives exactly $\begin{pmatrix} S & -T & -S \end{pmatrix}$.

An asymmetric port check. Take scalar w, z, u but a two-component measurement: $z = w + 2u$, $y = \begin{pmatrix}3\\5\end{pmatrix}w + \begin{pmatrix}1\\2\end{pmatrix}u$, $K = \begin{pmatrix}\frac{1}{10} & \frac{1}{5}\end{pmatrix}$. Then $u = 1.3w + 0.5u$, so $u = 2.6w$, $z = 6.2w$. Here $P_{22}K$ is 2×2, whereas $KP_{22} = 0.5$ is scalar. Both inverse forms give the same answer, using $K(I - P_{22}K)^{-1} = (I - KP_{22})^{-1}K$. One must preserve factor order: matrices cannot be commuted as in SISO arithmetic. The upper LFT, used later for uncertainty, closes the other ports: $\Delta \star P = P_{22} + P_{21}\Delta(I - P_{11}\Delta)^{-1}P_{12}$.

4 Stability is about the joint state, not one output

Definition (internal stability) [1, p. 51]. K stabilizes P if Equation 6 holds and

$$\mathcal{A} := \begin{pmatrix}A & 0\\ 0 & A_K\end{pmatrix} + \begin{pmatrix}B_2 & 0\\ 0 & B_K\end{pmatrix}\begin{pmatrix}I & -D_K\\ -D_{22} & I\end{pmatrix}^{-1}\begin{pmatrix}0 & C_K\\ C_2 & 0\end{pmatrix} \tag{12}$$

is Hurwitz. $\mathcal{A}$ is the state matrix of the closed loop in the joint state (x, x_K). Thus with $w = 0$, **every initial plant and controller state decays exponentially**. Transfer matrices instead describe zero-initial-state responses; cancelling a pole there does not remove stored energy from the physical system. Only (A, B_2, C_2, D_{22}) and the controller enter this matrix. For $D_{22} = 0$ it reduces to $\begin{pmatrix}A+B_2D_KC_2 & B_2C_K\\ B_KC_2 & A_K\end{pmatrix}$. The book's Example 3.4 applies this to the tracking loop with $G = 200/((10s+1)(0.05s+1)^2)$.

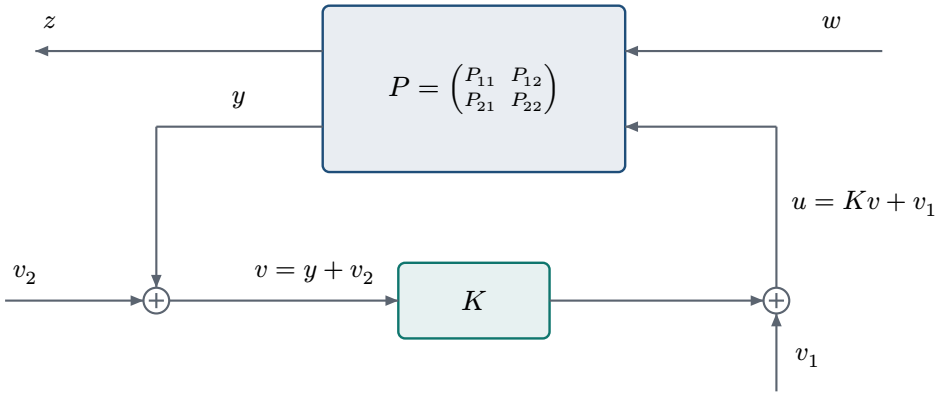


Figure 1: The closed loop $P \star K$ with test signals injected at both controller ports (after Scherer's Figure 22). Internal stability asks for a stable map from (w, v_1, v_2) to (z, u, v), not only from w to z.

5 Why $w \to z$ is not enough: expose the internal maps

$P \star K$ stable does not imply $\mathcal{A}$ Hurwitz: the closed-loop realization need not be stabilizable or detectable, so an unstable mode can be hidden from the channel $w \to z$. For a concrete asymmetric cancellation take $\dot{x}_1 = x_1 + w$, $\dot{x}_2 = x_2 + 2w$, $z = 2x_1 - x_2$. The zero-state map $w \to z$ is zero, but $x(0) = \begin{pmatrix}1\\0\end{pmatrix}$ gives $z(t) = 2\exp(t)$. The two unstable paths cancel only for a particular input pattern. The full open-loop realizations in the theorem below satisfy stabilizability/detectability; their **closed-loop performance channel** need not. Exercise 3b gives an example satisfying those open-loop assumptions.

Inject independent disturbances at the ports:

$$u = Kv + v_1, \quad v = y + v_2. \tag{13}$$

These test signals are probes, not additional controller information. With $S_y = (I - P_{22}K)^{-1}$ and $S_u = (I - KP_{22})^{-1}$, direct elimination gives

$$v = S_yP_{21}w + S_yP_{22}v_1 + S_yv_2, \tag{14}$$

$$u = KS_yP_{21}w + S_uv_1 + KS_yv_2. \tag{15}$$

The identity $I + KS_yP_{22} = S_u$ explains the direct $v_1 \to u$ term. Finally use $z = P_{11}w + P_{12}u$. This derives, rather than guesses, the nine maps.

Theorem 3.5 / Corollary 3.6 [1, pp. 55–57]. K stabilizes P iff $I - P_{22}K$ has a proper inverse and all nine blocks of

$$\begin{pmatrix} z \\ u \\ v \end{pmatrix} = \begin{pmatrix} P \star K & P_{12}(I - KP_{22})^{-1} & P_{12}K(I - P_{22}K)^{-1} \\ K(I - P_{22}K)^{-1}P_{21} & (I - KP_{22})^{-1} & K(I - P_{22}K)^{-1} \\ (I - P_{22}K)^{-1}P_{21} & (I - P_{22}K)^{-1}P_{22} & (I - P_{22}K)^{-1} \end{pmatrix} \begin{pmatrix} w \\ v_1 \\ v_2 \end{pmatrix} \tag{16}$$

are in $\mathcal{RH}_\infty$ (proper, real-rational, with poles strictly in the left half-plane). **Proof outline of the theorem:** the realization of the map $(w, v_1, v_2) \to (z, u, v)$ has state matrix exactly $\mathcal{A}$ of Equation 12, and it **is** stabilizable and detectable (Hautus test, using that both P and K are). For such realizations, a stable transfer matrix forces a Hurwitz state matrix.

For the tracking loop the lower-right 2×2 block is the familiar "gang of four" $(I + KG)^{-1}$, $K(I + GK)^{-1}$, $-(I + GK)^{-1}G$, $(I + GK)^{-1}$.

6 Generalized plants: when can P be stabilized at all?

Not every P admits a stabilizing K. In the book's Example 3.8, $P = \begin{pmatrix} 1 & 1/s \\ 1 & 1/(s+1) \end{pmatrix}$: the integrator sits in P_{12} but not in P_{22}, so feedback from y never "sees" it, and one of $P_{12}(1 - KP_{22})^{-1}$ or $P_{12}(1 - KP_{22})^{-1}K$ always keeps the pole at 0 [1, p. 58].

Definition. P is a **generalized plant** if at least one K stabilizes it [1, p. 59].

Theorem 3.10 (state-space test). With Equation 3 stabilizable and detectable, P is a generalized plant iff (A, B_2) is stabilizable and (A, C_2) is detectable. For the "if" part pick F, L with $A + B_2F$ and $A + LC_2$ Hurwitz; the observer-based controller

$$\dot{x}_K = (A + B_2F + LC_2 + LD_{22}F)x_K - Ly, \quad u = Fx_K \tag{17}$$

is obtained by estimating x from $y - D_{22}u$: choose $\dot{x}_K = Ax_K + B_2u - L(y - C_2x_K - D_{22}u)$ and $u = Fx_K$. At $w = 0$, put $e = x - x_K$. Direct subtraction gives

$$\dot{x} = (A + B_2F)x - B_2Fe, \quad \dot{e} = (A + LC_2)e. \tag{18}$$

Thus $\mathcal{A} \cong \begin{pmatrix} A+B_2F & -B_2F \\ 0 & A+LC_2 \end{pmatrix}$. The off-diagonal minus sign follows from $x_K = x - e$; only the two Hurwitz diagonal blocks determine the eigenvalues. Since $D_K = 0$, well-posedness is automatic. For "only if": if $Ax = \lambda x$, $C_2x = 0$, then $\mathcal{A}\begin{pmatrix} x \\ 0 \end{pmatrix} = \lambda\begin{pmatrix} x \\ 0 \end{pmatrix}$, so $\operatorname{Re}\lambda < 0$; (A, C_2) is detectable.

> The book leaves stabilizability of (A, B_2) as a task. Prove it in the same way.

Take a left eigenvector: $x^*A = \lambda x^*$, $x^*B_2 = 0$, $x \neq 0$. In Equation 12 the row vector $\begin{pmatrix} x^* & 0 \end{pmatrix}$ annihilates the factor $\begin{pmatrix} B_2 & 0 \\ 0 & B_K \end{pmatrix}$, hence $\begin{pmatrix} x^* & 0 \end{pmatrix}\mathcal{A} = \begin{pmatrix} x^*A & 0 \end{pmatrix} = \lambda\begin{pmatrix} x^* & 0 \end{pmatrix}$. So λ is an eigenvalue of the Hurwitz matrix $\mathcal{A}$ and $\operatorname{Re}\lambda < 0$: every uncontrollable mode of (A, B_2) is stable.

Theorem 3.12 (input-output test) [1, pp. 60–61]. Let K be **any** controller that stabilizes $y = P_{22}u$ (one always exists). Then P is a generalized plant iff this K stabilizes P, i.e. renders all nine blocks of Equation 16 stable. One candidate K thus decides the question. Here "stabilizes P_{22}" refers to a minimal (or stabilizable/detectable) realization of that transfer matrix, not the possibly nonminimal realization inherited from all of P. Otherwise the assertion that a stabilizer always exists would be false.

Why this test works (outline). If P is a generalized plant, the state-space test makes (A, B_2, C_2, D_{22}) stabilizable/detectable. No unstable mode can then disappear when it is reduced to a minimal realization of P_{22}. A controller stabilizing this channel stabilizes the same unstable modes of the whole system. Conversely, passing all nine tests is already the definition's input-output characterization.

Existence here allows **unrestricted** dynamic output feedback. It does not promise diagonal, low-order, static, or stable controllers; structural constraints require additional work, as Exercise 3c demonstrates.

Tracking loop revisited [1, pp. 62–63]. With a minimal realization (A_G, B_G, C_G, D_G) of G, Equation 1 has $A = A_G$, $B_2 = B_G$, $C_2 = -C_G$; controllability and observability of G make it a generalized plant. In input-output form: any K that stabilizes $-G$ makes all nine blocks stable, because P_{11}, P_{21} are constant and $P_{12} = G$ only appears in $G(I + KG)^{-1}$ and $GK(I + GK)^{-1} = I - (I + GK)^{-1}$.

Workflow. Extract P; test the generalized-plant property (state space or one stabilizing K for P_{22}); design K; closed loop $= P \star K$ with realization built from Equation 12.

7 Exercise 1: stable P and the Youla parameter

> **Exercise 1.** Let P be stable. (a) Show K stabilizes P iff $I - P_{22}K$ has a proper inverse and $K(I - P_{22}K)^{-1}$ is stable. Is the same true with $(I - P_{22}K)^{-1}$ instead? (b) Show that $\{P \star K : K \text{ stabilizes } P\}$ equals the set of all $P_{11} + P_{12}QP_{21}$, $Q \in \mathcal{RH}_\infty$, and relate K and Q.

(a) Assume $I - P_{22}K$ has a proper inverse and put

$$Q := K(I - P_{22}K)^{-1} = (I - KP_{22})^{-1}K, \tag{19}$$

the second form by the push-through identity. Then

$$(I - P_{22}K)^{-1} = I + P_{22}Q, \quad (I - KP_{22})^{-1} = I + QP_{22}. \tag{20}$$

Indeed $(I - P_{22}K)^{-1} - I = (I - P_{22}K)^{-1}P_{22}K = P_{22}(I - KP_{22})^{-1}K = P_{22}Q$, and symmetrically $(I - KP_{22})^{-1} - I = K(I - P_{22}K)^{-1}P_{22} = QP_{22}$. Substituting Equation 19–Equation 20 into Equation 16, the nine blocks become

$$\begin{pmatrix} P_{11} + P_{12}QP_{21} & P_{12}(I + QP_{22}) & P_{12}Q \\ QP_{21} & I + QP_{22} & Q \\ (I + P_{22}Q)P_{21} & (I + P_{22}Q)P_{22} & I + P_{22}Q \end{pmatrix}. \tag{21}$$

Every entry is a sum of products of blocks of P and Q. If $Q \in \mathcal{RH}_\infty$ and $P \in \mathcal{RH}_\infty$ they are all stable, so K stabilizes P by Corollary 3.6. Conversely, if K stabilizes P, then Q is the $(2,3)$ block and is stable. One test replaces nine.

With $(I - P_{22}K)^{-1}$ instead: false. Take $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ (stable, $P_{22} = 0$) and $K = 1/(s-1)$. Then $(I - P_{22}K)^{-1} = 1$ is stable, but $K(I - P_{22}K)^{-1} = K$ and even $P \star K = K$ are unstable. The reason: $(I - P_{22}K)^{-1} = I + P_{22}Q$ only sees Q through P_{22}, which can annihilate (here: entirely) the unstable part of Q.

(b) "$\subset$": if K stabilizes P, Q from Equation 19 lies in $\mathcal{RH}_\infty$ and $P \star K = P_{11} + P_{12}QP_{21}$ by definition Equation 11. Moreover $I + P_{22}Q = (I - P_{22}K)^{-1}$ has the proper inverse $I - P_{22}K$.

"$\supset$": given $Q \in \mathcal{RH}_\infty$ with $I + P_{22}(\infty)Q(\infty)$ nonsingular, define

$$K := Q(I + P_{22}Q)^{-1}, \tag{22}$$

which is proper. Then $I - P_{22}K = (I + P_{22}Q - P_{22}Q)(I + P_{22}Q)^{-1} = (I + P_{22}Q)^{-1}$, whose inverse $I + P_{22}Q$ is proper, and $K(I - P_{22}K)^{-1} = Q(I + P_{22}Q)^{-1}(I + P_{22}Q) = Q \in \mathcal{RH}_\infty$. By (a), K stabilizes P, and $P \star K = P_{11} + P_{12}QP_{21}$.

The maps Equation 19 and Equation 22 are mutual inverses, so $K \leftrightarrow Q$ is a bijection between stabilizing controllers and admissible Youla parameters, and the closed loop is **affine** in Q.

Caveat on "free" Q. The side condition $\det(I + P_{22}(\infty)Q(\infty)) \neq 0$ is genuinely needed when $D_{22} \neq 0$: for the static $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ one has $P \star K = K/(1+K)$, which never equals $1 = P_{11} + P_{12} \cdot 1 \cdot P_{21}$ (the value for $Q = 1$). When P_{22} is strictly proper, the most common case, the condition is void and Q ranges over all of $\mathcal{RH}_\infty$.

8 Exercise 2: which P are generalized plants?

> **Exercise 2.** All signals scalar. Which of
>
> $$P_a = \begin{pmatrix} \frac{1}{s+1} & \frac{1}{s+2} \\ \frac{1}{s+3} & \frac{1}{s} \end{pmatrix}, \quad P_b = \begin{pmatrix} \frac{1}{s^2+s} & \frac{1}{s+2} \\ \frac{1}{s+3} & \frac{1}{s} \end{pmatrix}, \tag{23}$$
>
> $$P_c = \begin{pmatrix} \frac{1}{s+1} & \frac{1}{s^2+2s} \\ \frac{1}{s+3} & \frac{1}{s} \end{pmatrix}, \quad P_d = \begin{pmatrix} \frac{1}{s+1} & \frac{1}{s+2} \\ \frac{1}{s+3} & \frac{1}{s^2} \end{pmatrix} \tag{24}$$
>
> define a generalized plant?

Use Theorem 3.12: stabilize P_{22} with some K, then check Equation 16. All P_{22} are strictly proper, so well-posedness is automatic.

P_a, P_b, P_c ($P_{22} = 1/s$). $K = -1$ stabilizes $1/s$: $S := (1 - P_{22}K)^{-1} = s/(s+1)$, $KS = -s/(s+1)$, $SP_{22} = 1/(s+1)$, all stable (scalars, so $(1 - KP_{22})^{-1} = S$). The remaining blocks are

$$P \star K = P_{11} - P_{12}P_{21}\frac{s}{s+1}, \quad P_{12}S, \quad P_{12}KS, \quad KSP_{21}, \quad SP_{21}. \tag{25}$$

- P_a: P_{11}, P_{12}, P_{21} stable, so all of Equation 25 is stable. **Generalized plant.**
- P_b: P_{12}, P_{21} are stable, but $P \star K = 1/(s(s+1)) - s/((s+1)(s+2)(s+3))$ keeps the pole at 0 from P_{11}. **Not a generalized plant.** The integrator lives in the channel $w \to z$ as well as in $u \to y$; the residue of P_b at 0 is $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ (rank 2), so a minimal realization has **two** integrator states, and only one of them is reachable from u.
- P_c: $P_{12} = 1/(s(s+2))$ is unstable, but $P_{12}S = 1/((s+1)(s+2))$, $P_{12}KS = -1/((s+1)(s+2))$ and $P \star K = 1/(s+1) - 1/((s+1)(s+2)(s+3))$ are stable; so are the others. **Generalized plant.** Here the residue at 0 is $\left(\begin{smallmatrix} 0 & 1/2 \\ 0 & 1 \end{smallmatrix}\right)$ (rank 1): one integrator, reachable from u and visible in y.

P_d ($P_{22} = 1/s^2$). $K = -1$ fails ($s^2 + 1$). Take the lead controller $K = -(3s+1)/(s+3)$: $1 - P_{22}K = (s^3 + 3s^2 + 3s + 1)/(s^2(s+3)) = (s+1)^3/(s^2(s+3))$, so $S = s^2(s+3)/(s+1)^3$, $KS = -s^2(3s+1)/(s+1)^3$, $SP_{22} = (s+3)/(s+1)^3$, all stable. P_{11}, P_{12}, P_{21} are stable, so every block of Equation 16 is a product of stable factors. **Generalized plant.**

Answer: P_a, P_c, P_d are generalized plants; P_b is not. Rule of thumb: every unstable pole of P must appear in P_{22} with its full multiplicity (McMillan degree), so that it is both reachable from u and visible in y.

9 Exercise 3: the interconnection of Figure 23

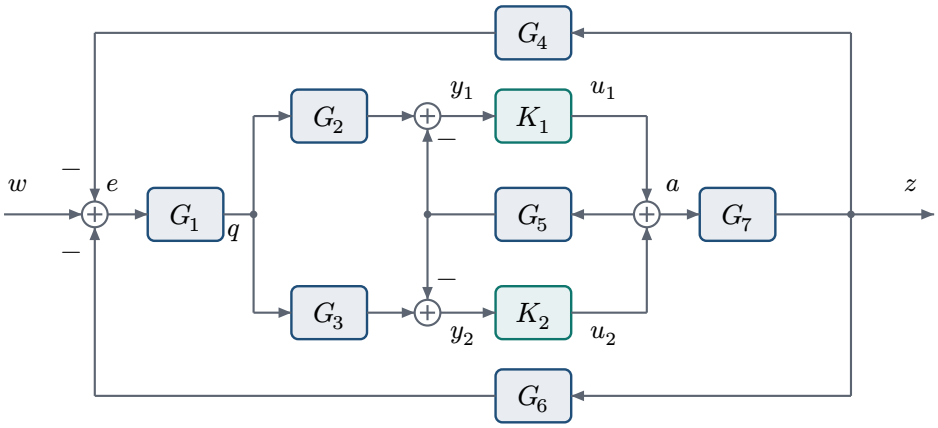


Figure 2: Scherer's Figure 23, redrawn. Controller blocks in teal; the summing junction after K_1, K_2 produces $a = u_1 + u_2$, which feeds both G_7 and G_5. Signs at the left and middle junctions are minus where marked.

> **Exercise 3.** For Figure 2, with possibly MIMO G_j and K_j: (a) compute P in terms of G_j; (b) find SISO examples P_1 that is no generalized plant, and P_2 with a K that makes $P_2 \star K$ stable without stabilizing P_2; can $P_1 = P_2$? (c) for the data below show P is a generalized plant, design a stabilizing K, explain the check, and plot $|P \star K|$.

(a) Open-loop description. Cut the controller: inputs w, $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$; outputs z, $y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$; the controller is block diagonal, $K = \operatorname{diag}(K_1, K_2)$. Read off the diagram, keeping the order of factors:

$$a = u_1 + u_2, \quad z = G_7 a, \quad e = w - (G_4 + G_6) z, \quad q = G_1 e, \tag{26}$$

$$y_1 = G_2 q - G_5 a, \quad y_2 = G_3 q - G_5 a. \tag{27}$$

There is no algebraic loop once u is fixed: z depends on u only. Substituting, $q = G_1 w - G_1(G_4 + G_6) G_7 (u_1 + u_2)$, and with

$$H_j := G_j G_1 (G_4 + G_6) G_7 + G_5, \quad j = 2, 3, \tag{28}$$

$$\begin{pmatrix} z \\ y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 & G_7 & G_7 \\ G_2 G_1 & -H_2 & -H_2 \\ G_3 G_1 & -H_3 & -H_3 \end{pmatrix} \begin{pmatrix} w \\ u_1 \\ u_2 \end{pmatrix}. \tag{29}$$

Dimension constraints implied by the diagram: u_1, u_2 and a share one dimension; y_1 and y_2 must have equal dimension, since the same $G_5 a$ is subtracted from both; G_4, G_6 map z into the dimension of w. Note $P_{11} = 0$ and $P_{12} = \begin{pmatrix} G_7 & G_7 \end{pmatrix}$: both control inputs act through the same channel. The numerical state-space build for part (c) agrees with this formula to 2.4×10^{-15} at three complex test points.

(b) Examples. P_1, no generalized plant: $G_1 = G_3 = G_7 = 1$, $G_2 = 1/(s-1)$, $G_4 = G_5 = G_6 = 0$. Then

$$P_1 = \begin{pmatrix} 0 & 1 & 1 \\ \frac{1}{s-1} & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \tag{30}$$

Since $P_{22} = 0$, the block $(I - P_{22}K)^{-1} P_{21} = P_{21}$ of Equation 16 contains $1/(s-1)$ for **every** K: the unstable G_2 is driven by w and cannot be influenced by u because $G_4 = G_6 = 0$ cut the path from u back to G_2.

P_2, stable $P_2 \star K$ without stabilization: $G_1 = G_2 = G_3 = G_7 = 1$, $G_4 = G_5 = G_6 = 0$, so $P_2 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$, which is stable and hence a generalized plant ($K = 0$ stabilizes it). Take $K_1 = 1/(s-1)$, $K_2 = -1/(s-1)$. Then $P_2 \star K = \begin{pmatrix} 1 & 1 \end{pmatrix} \operatorname{diag}(K_1, K_2) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = K_1 + K_2 = 0$, stable, yet $K(I - P_{22}K)^{-1} = K$ is unstable, so K does not stabilize P_2 (Exercise 1a). The unstable controller modes cancel at the junction producing a, but u_1 and u_2 individually grow without bound.

Can $P_1 = P_2$? Yes. For P_1 above and $K = 0$, $P_1 \star K = P_{11} = 0$ is stable, while no controller at all stabilizes P_1. So the same interconnection serves both roles, which shows sharply that a stable $w \to z$ map certifies nothing.

(c) Data. $G_1 = 1$, $G_2 = 1/(s-1)$, $G_3 = (s+1)/(s^2+1)$, $G_4 = 0$, $G_5 = 1/s$, $G_6 = 1$, $G_7 = (s+2)/((s+3)(s-2))$, so $H_j = G_j G_7 + 1/s$ in Equation 28.

Generalized plant. Build a realization Equation 3 from realizations of G_7 (2 states), G_2 (1), G_3 (2), G_5 (1), with $x = (x_7, x_2, x_3, x_5)$: $\dot{x}_7 = A_7 x_7 + B_7 a$, $\dot{x}_2 = A_2 x_2 + B_2^G (w - C_7 x_7)$, likewise x_3, $\dot{x}_5 = a$, and $B_2 = \begin{pmatrix} b_a & b_a \end{pmatrix}$ with b_a the input column of a. All blocks are strictly proper, so $D_{22} = 0$. The eigenvalues of A are $1, 2, \pm i, 0, -3$. Numerically, $(A, \begin{pmatrix} B_1 & B_2 \end{pmatrix})$ is controllable and $\left(A, \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}\right)$ observable (ranks $6 = n$), so the realization is minimal, and the Hautus matrices $\begin{pmatrix} A - \lambda I & B_2 \end{pmatrix}$ and $\begin{pmatrix} A - \lambda I \\ C_2 \end{pmatrix}$ have rank 6 at each of $\lambda = 1, 2, \pm i, 0$. By Theorem 3.10, P is a generalized plant. The structure explains why: a reaches every mode (directly for G_7, G_5; via $z = G_7 a$ for G_2, G_3), the G_2 mode is seen in y_1, the G_3 modes in y_2, and the integrator in both.

Design. Theorem 3.10′s observer controller Equation 17 applied to the 2-input, 2-output P gives a **full** 2×2 K, which stabilizes P in the sense of the definition but cannot be built in Figure 2: there K_1 sees only y_1 and K_2 only y_2. Realizing separate copies of a centralized controller can introduce hidden unstable duplicate states; it is not a valid shortcut to decentralized stabilization. We therefore design $K = \mathrm{diag}(K_1, K_2)$ directly, in two steps:

1. Choose $K_2 = 1$ (static, so K_1 and K_2 share no poles). Closing $u_2 = y_2$ gives $A' = A + b_a C_{2,2}$, where $C_{2,2}$ is the y_2 row of C_2. It is needed: with $K_2 = 0$ the modes $\pm i$ of G_3 would be invisible to K_1.
2. For the SISO loop $\left(A', b_a, C_{2,1}\right)$, which is controllable and observable (ranks 6), apply Equation 17 with LQR gains (weights I and 1) for F and the dual problem for L: $\dot{\xi} = \left(A' + b_a F + L C_{2,1}\right)\xi - L y_1, u_1 = F\xi$.

The resulting sixth-order controller is

$$K_1(s) = \frac{479.08 s^5 + 1557.5 s^4 + 722.32 s^3 + 1247.6 s^2 + 977.12 s - 626.62}{s^6 + 15.301 s^5 - 371.51 s^4 - 1087.4 s^3 + 272.67 s^2 - 1079.7 s + 641.85}, \tag{31}$$

with poles -27.40, 14.77, -3.17, 0.506, $-0.0043 \pm 0.994i$. An unstable controller is allowed; only internal stability matters.

Check. (i) Equation 6: $D_{22} = 0$, so it holds (det $= 1$). (ii) $\mathcal{A}$ from Equation 12, 12×12, has eigenvalues $-3.076 \pm 0.862i$, -2.898, $-0.338 \pm 1.129i$, $-0.190 \pm 1.029i$, $-1.842 \pm 0.456i$, -0.734, -0.743, -1.034: largest real part -0.190, so K stabilizes P. (iii) Physical substitution $a = F\xi + y_2$, $\dot{\xi} = A_1\xi - L y_1$ reproduces the full state matrix exactly; the error-coordinate triangularization has residual 8.9×10^{-16}. (iv) The full disturbed port map agrees with direct solution of its signal equations to 4.1×10^{-11} at three complex points. Its joint minimal realization has 12 stable poles; this is not a claim that each individual block has all 12 poles. (v) The realization of $P \star K$ matches Equation 11 evaluated pointwise (error 2.2×10^{-12}).

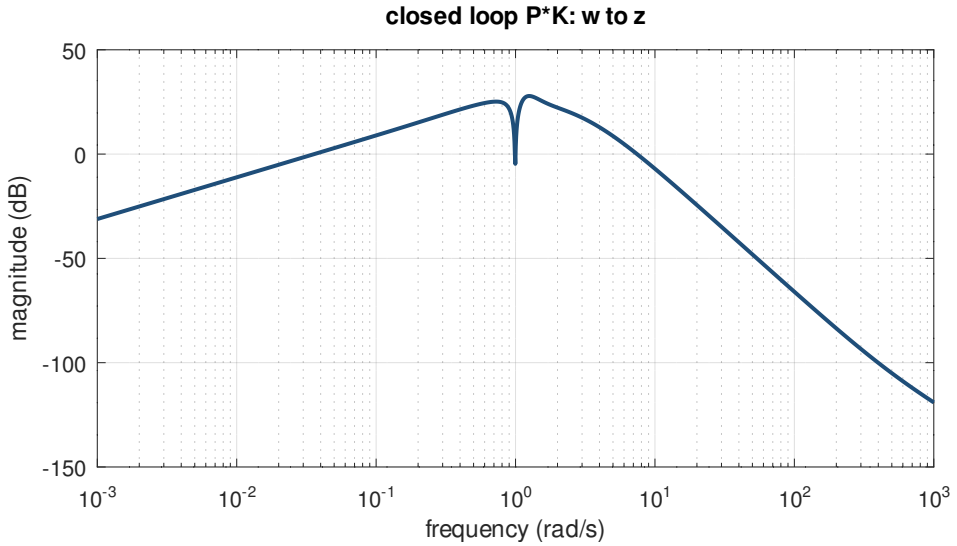


Figure 3: Bode magnitude of the closed loop $P \star K$ ($w \to z$) for the design of Exercise 3c. Peak 24.66 (27.8 dB) at $\omega \approx 1.25$ rad/s; zero DC gain; roll-off -40 dB/decade.

Reading the plot. $P \star K(0) = 0$ for **every** stabilizing K: internal stability makes the maps from w to y_1, $q = w - z$ and $a = u_1 + u_2$ stable, and $s^{-1}a = G_5 a = G_2 q - y_1$. Since G_2 is analytic at 0, the right side is bounded near $s = 0$, so the map $w \to a$ must vanish at $s = 0$, and so does $z = G_7 a$. The large resonant peak near $\omega \approx 1$ (close to the G_3 poles $\pm i$ and to the lightly damped closed-loop pair $-0.19 \pm 1.03i$) shows that this design only achieves stability; shaping $|P \star K|$ is the subject of the performance chapters.

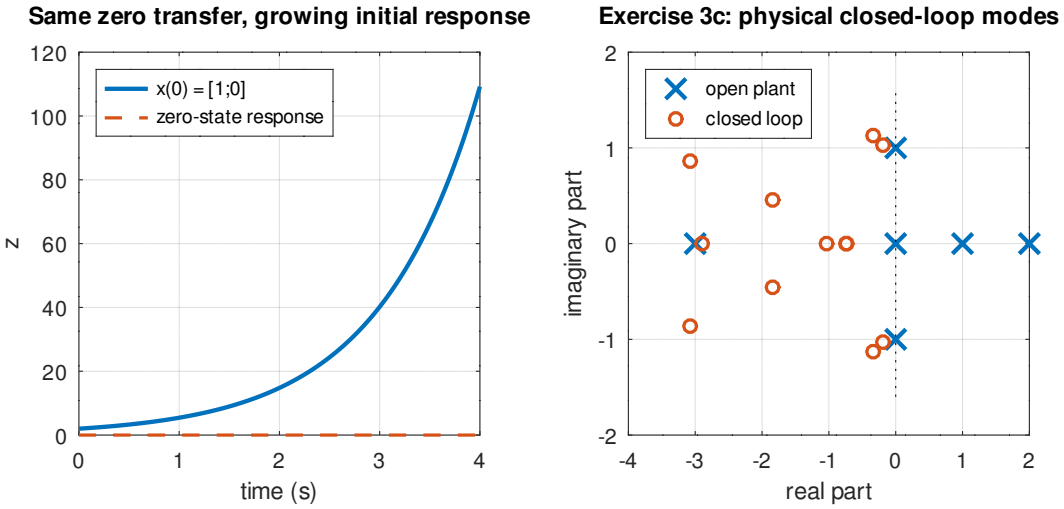


Figure 4: Two computed diagnostics. Left: the asymmetric cancellation example has zero zero-state output but an exponentially growing initial-state response. Right: Exercise 3c moves the physical joint-state modes into the left half-plane. A magnitude plot alone cannot supply either initial-condition or hidden-mode information.

Reproduce. From the repository root: `octave --no-gui Control/robust-control-03-interconnections/scripts/ex3c.m` (GNU Octave 11.3 with the `control` package; it prints every number above and writes both SVG plots). Use the state-space construction, not the rounded polynomial above, as the reproducible controller specification. The script replaces MATLAB's `sysic` by an explicit state-space build and uses `lqr` for F and L, so it runs without the Robust Control Toolbox. With identity LQR state weights, even the designed transfer controller depends on the realization basis. A coordinate change preserves the design only if the weights are transformed consistently. The supplied script fixes the basis and weights together.

10 Sources and further reading

All theory statements, equation forms, and exercise statements follow Chapter 3 of [1] (pp. 46–65), read from the PDF. The exercise solutions, counterexamples, the decentralized design in Exercise 3c, and the remark on the Youla side condition are derived here, with reproducible Octave numerical checks (see `summary.md`). The nine-map theorem and input-output existence theorem have proof outlines, not full Hautus-test proofs. No other sources were consulted.

Bibliography

[1] C. W. Scherer, "Theory of Robust Control." [Online]. Available: https://www.imng.uni-stuttgart.de/mst/files/RC.pdf